

MERN STACK INTERNSHIP – DAY 20 ASSIGNMENT REPORT

Company

Insta Dot Analytics

Submitted By

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Submission Date: 27-06-2026

FOR SOURCE CODE:

GIT HUB :- <https://github.com/chayan-soni/MERN-Internship>

MERN Stack Internship Assignment Report

Project Title

Online Learning Management System (LMS)

Objective

The objective of this project is to design and develop a complete Online Learning Management System (LMS) using the MERN Stack. The application provides a secure platform where instructors can create and manage courses, while students can enroll in courses, submit assignments, attempt quizzes, and receive certificates upon successful completion. The project also focuses on implementing modern authentication mechanisms, responsive user interfaces, efficient state management, and deployment on cloud platforms.

Technologies Used

Frontend

- React.js

- Redux Toolkit
- React Router DOM
- Axios
- Tailwind CSS / CSS
- Chart.js (Dashboard Charts)

Backend

- Node.js
- Express.js
- MongoDB
- Mongoose
- JWT Authentication
- Multer (File Upload)
- Nodemailer (Email Services)

Deployment

- Frontend: Vercel
- Backend: Render

Authentication Module

The application implements a secure authentication system using JSON Web Tokens (JWT). User passwords are securely stored after encryption, and authenticated users receive JWT access tokens for protected resources. A Refresh Token mechanism is implemented to maintain secure sessions without requiring frequent logins.

Authentication Features

- User Registration
- User Login
- JWT Authentication
- Refresh Token Mechanism
- Protected Routes
- Forgot Password
- Email Verification
- Profile Management

Project Modules

Student Module

- Student Registration and Login
- Course Enrollment
- Assignment Submission
- Quiz Participation
- Certificate Download
- Dashboard Access
- Profile Management

Instructor Module

- Instructor Authentication
- Course Management
- Assignment Management
- Quiz Creation
- Student Progress Monitoring

Course Module

- Create Course
- Update Course
- Delete Course
- View Courses
- Search Courses
- Filter Courses
- Pagination Support

Assignment Module

- Upload Assignments
- Assignment Submission
- File Upload
- Submission Tracking

Quiz Module

- Quiz Creation
- Quiz Timer
- Automatic Evaluation

- Result Generation

Certificate Module

- Automatic Certificate Generation
- Certificate Download after Course Completion

Functionalities Implemented

1. Course CRUD

The instructor can create, read, update, and delete courses using REST APIs integrated with MongoDB.

2. Student Enrollment

Students can browse available courses and enroll in their preferred courses.

3. Assignment Submission

Students can upload assignments securely using file upload functionality.

4. Quiz Timer

Time-based quizzes automatically submit when the timer expires.

5. Certificate Generation

Certificates are generated automatically after successful completion of the course.

6. Dashboard Charts

Interactive charts display statistics such as enrolled courses, completed courses, assignments, and quiz performance.

7. Search

Users can search courses quickly using keywords.

8. Filter

Courses can be filtered according to category or instructor.

9. Pagination

Pagination improves performance by displaying limited records per page.

10. File Upload

Secure file upload functionality is implemented for assignments and profile images.

11. Profile Management

Users can update their personal information and profile image.

12. Forgot Password

Password reset functionality is provided through email verification.

13. Email Verification

Users verify their email address during registration to improve security.

14. Redux Toolkit

Redux Toolkit is used for centralized state management across the application.

15. Responsive UI

The application is fully responsive and works efficiently on desktop, tablet, and mobile devices.

16. Frontend Deployment

The frontend application is deployed using Vercel for fast and reliable hosting.

17. Backend Deployment

The backend APIs are deployed using Render with proper environment variable configuration.

API Modules

- Authentication APIs
- Student APIs
- Instructor APIs

- Course APIs
- Assignment APIs
- Quiz APIs
- Certificate APIs
- Profile APIs

All APIs are developed using Express.js and tested using Postman.

Database Design

MongoDB is used as the primary database. Mongoose ODM manages communication between the application and the database. Collections include:

- Users
- Courses
- Enrollments
- Assignments
- Quizzes
- Certificates

Security Features

- JWT Authentication
- Refresh Token Mechanism
- Protected Routes
- Password Encryption
- Email Verification
- Input Validation
- Secure API Access
- Environment Variables for Sensitive Credentials

Learning Outcomes

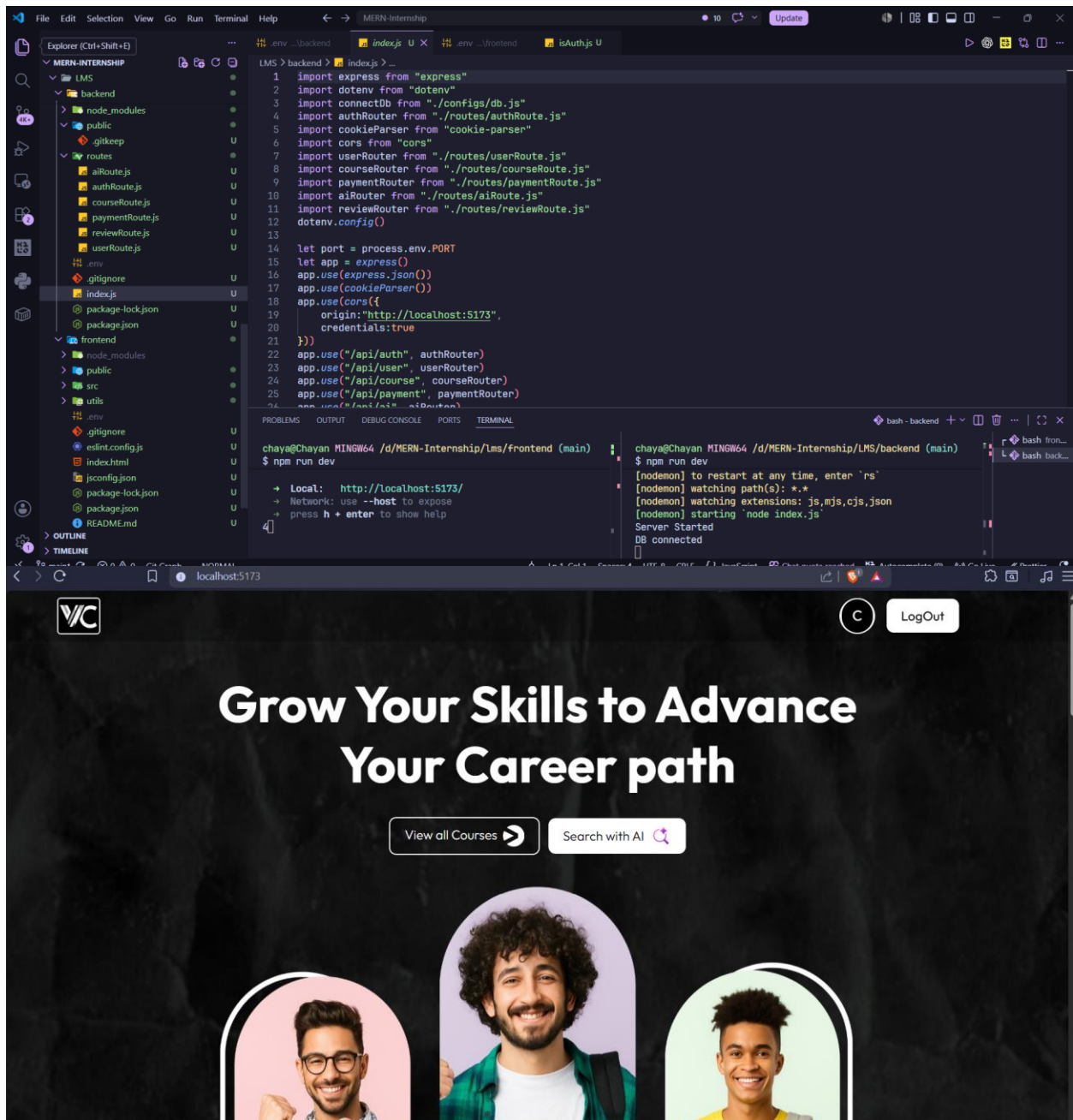
Through this project, the following concepts were learned and implemented:

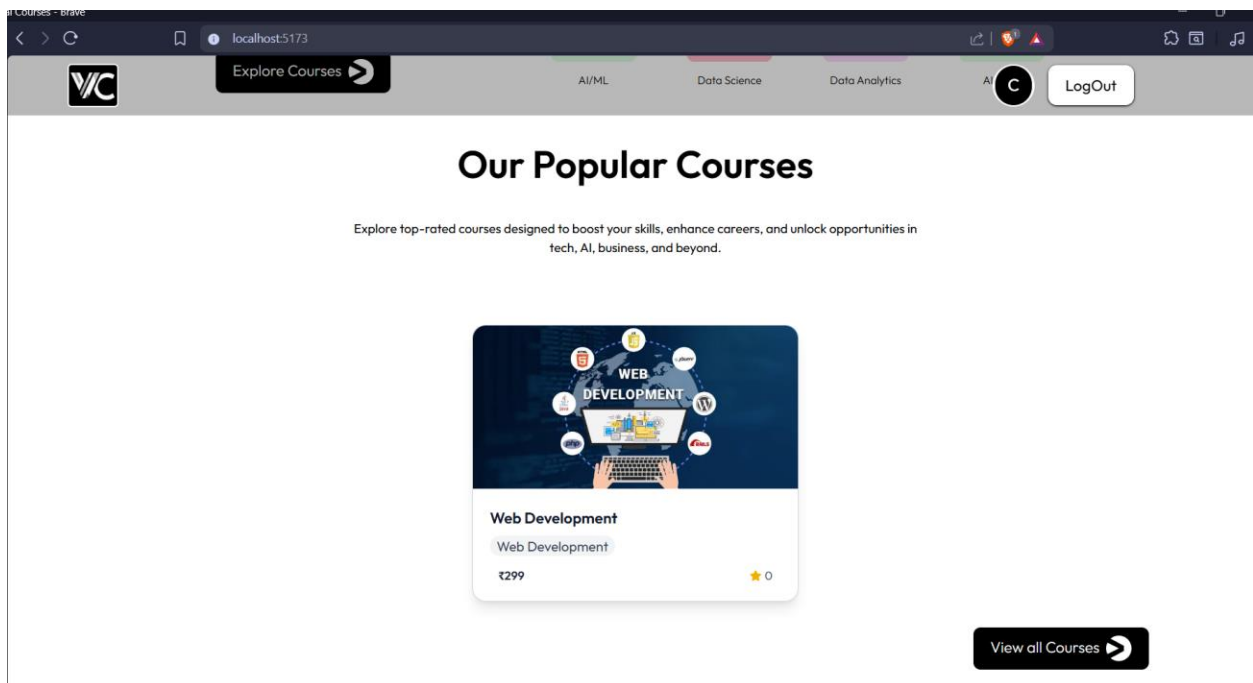
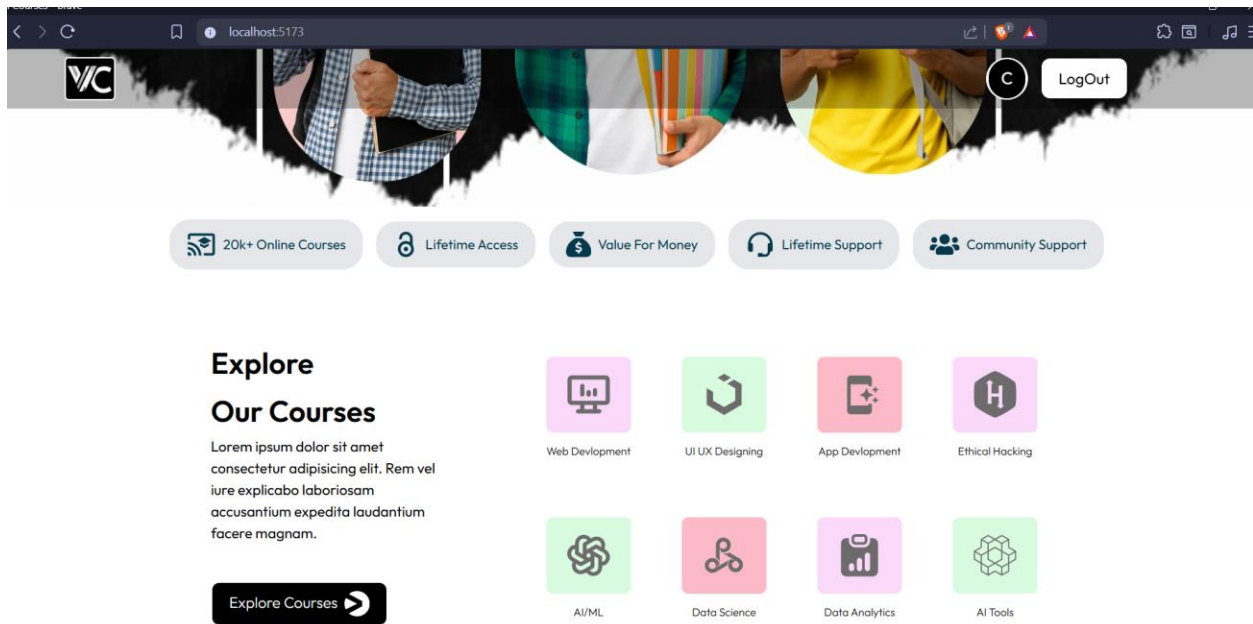
- Complete MERN Stack Development
- REST API Development

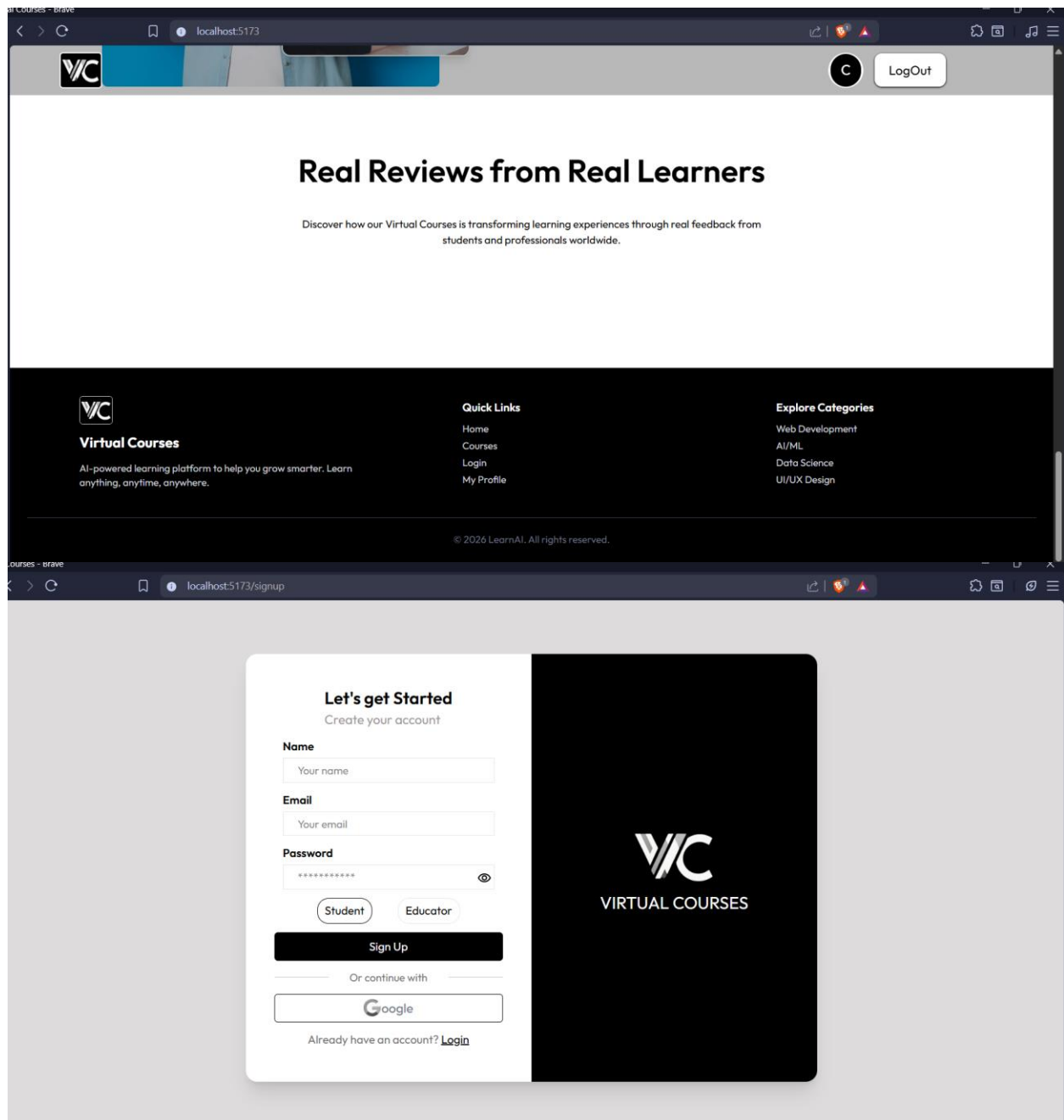
- MongoDB Database Integration
- JWT Authentication
- Refresh Token Authentication
- Redux Toolkit State Management
- Protected Routes
- File Upload using Multer
- Email Verification
- Responsive Web Design
- Dashboard Visualization
- Frontend and Backend Deployment
- Project Structure and Best Coding Practices

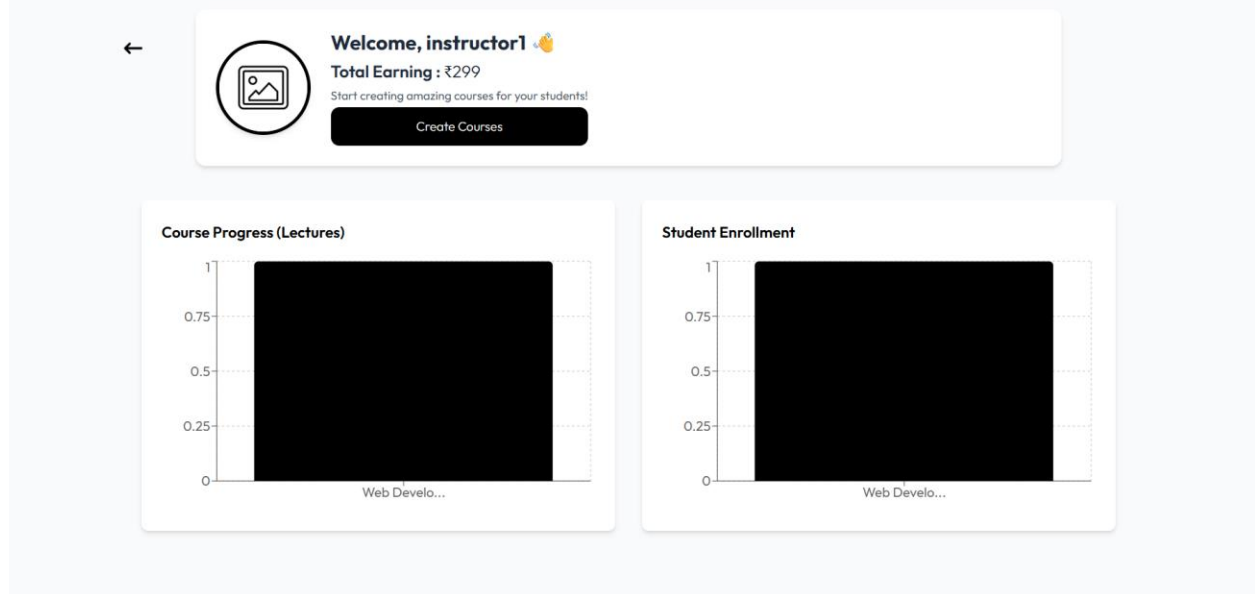
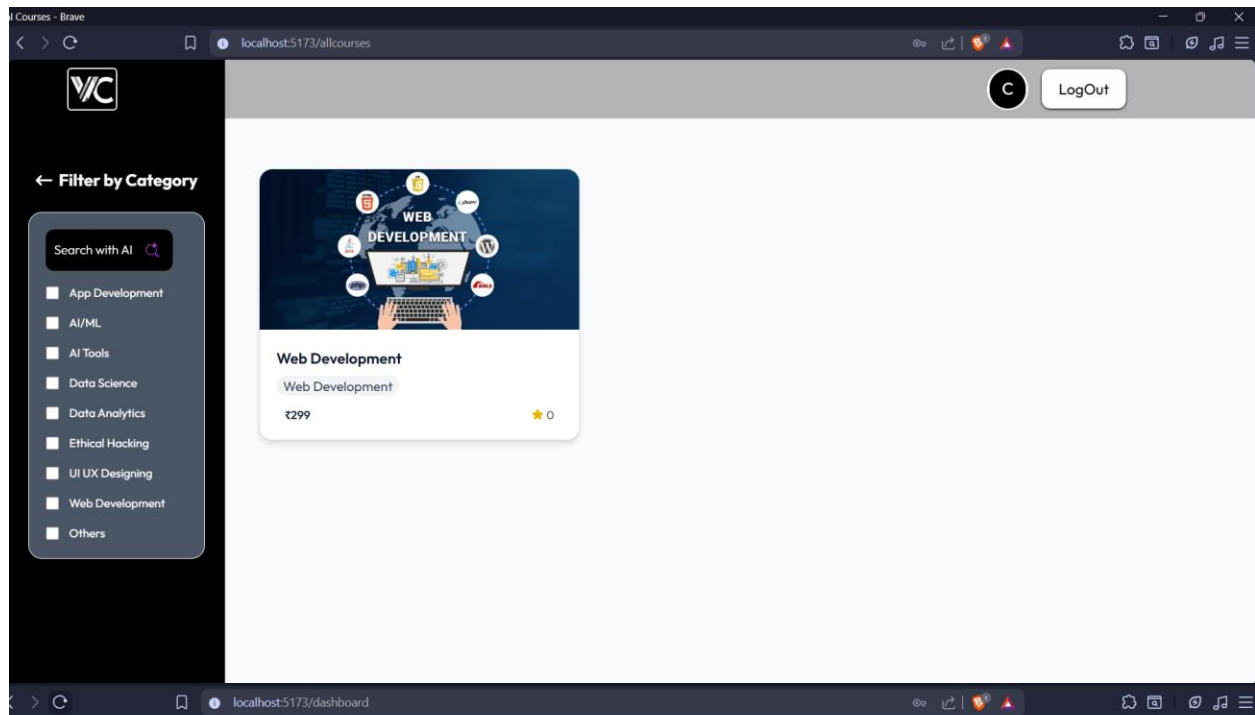
Conclusion

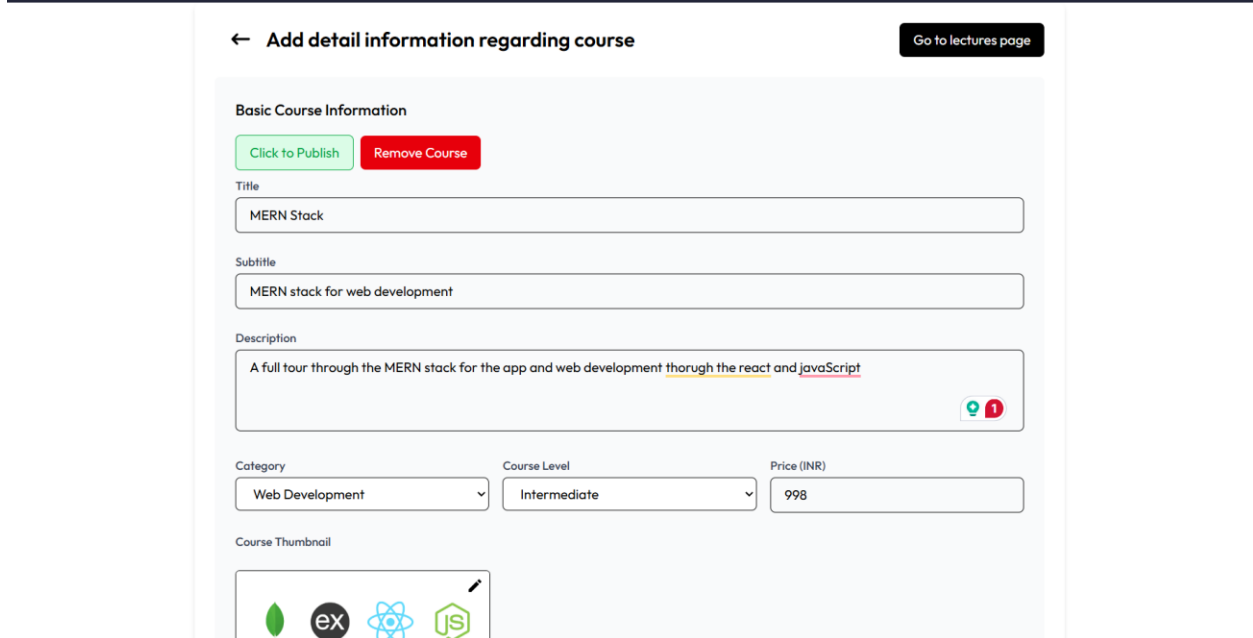
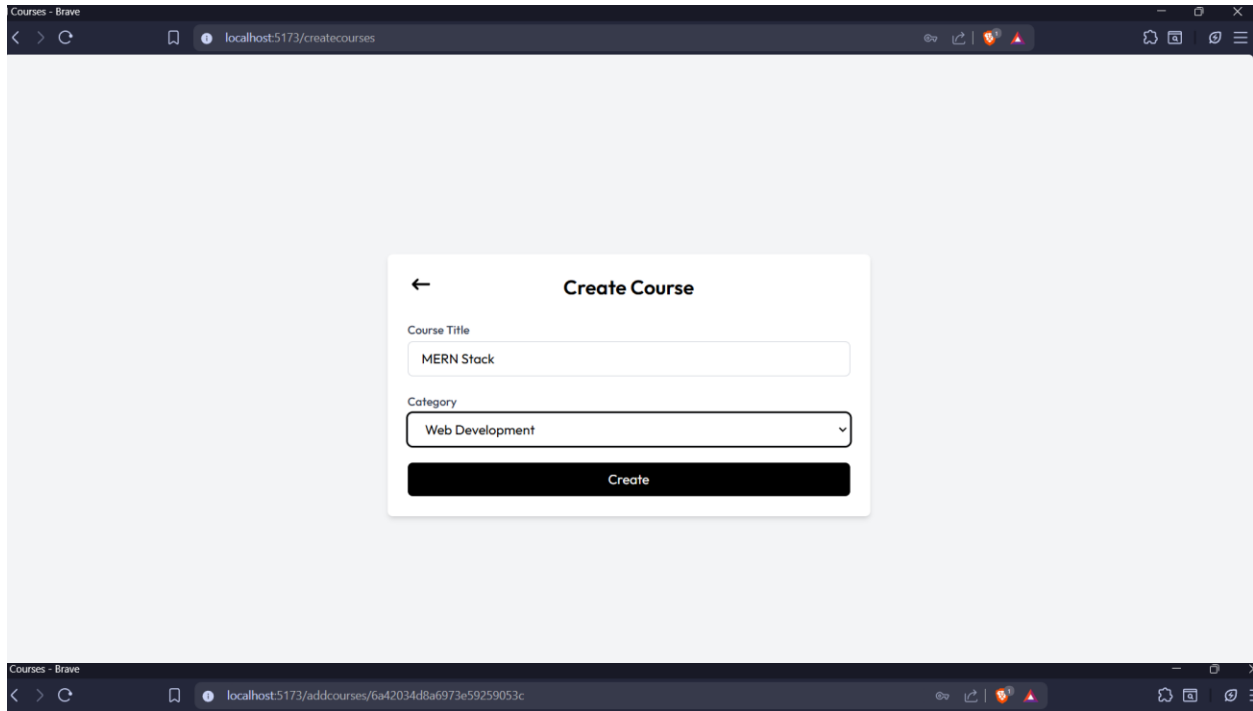
The Online Learning Management System successfully demonstrates the implementation of a full-stack web application using the MERN Stack. The project integrates secure authentication, course management, assignment submission, quizzes, certificate generation, dashboard analytics, responsive user interface, and cloud deployment. This project strengthened practical knowledge of modern web development, backend API design, database management, authentication mechanisms, and deployment workflows, making it a comprehensive real-world learning experience.

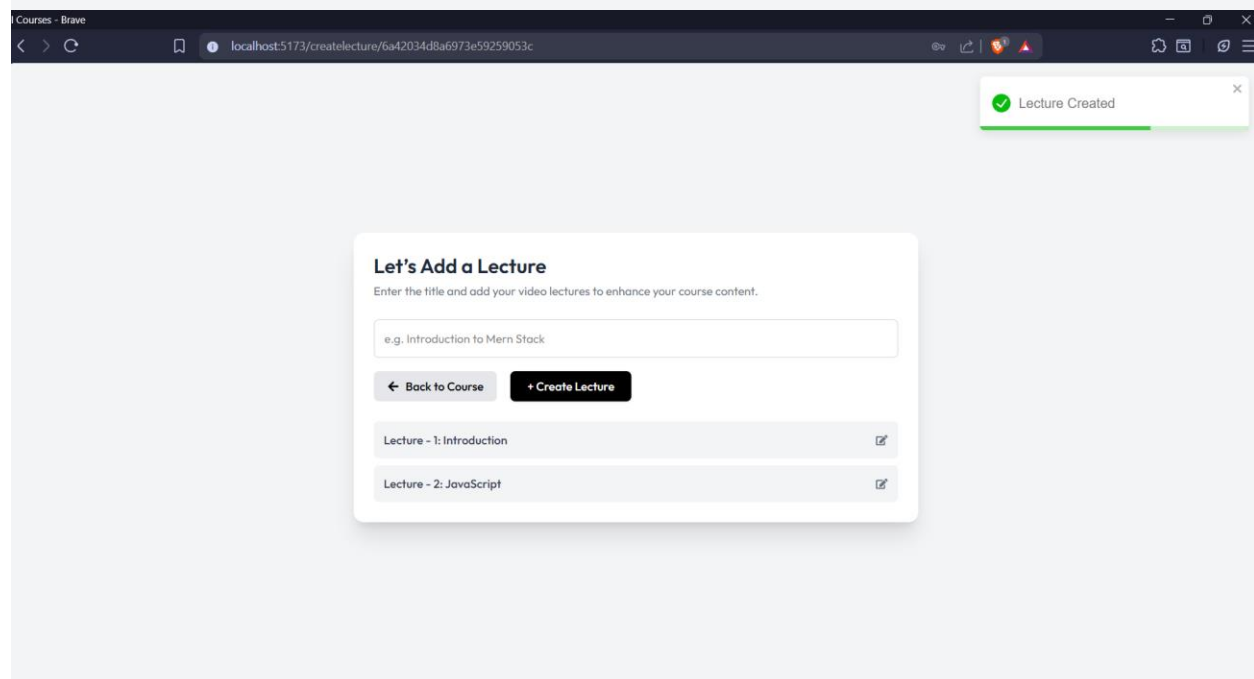
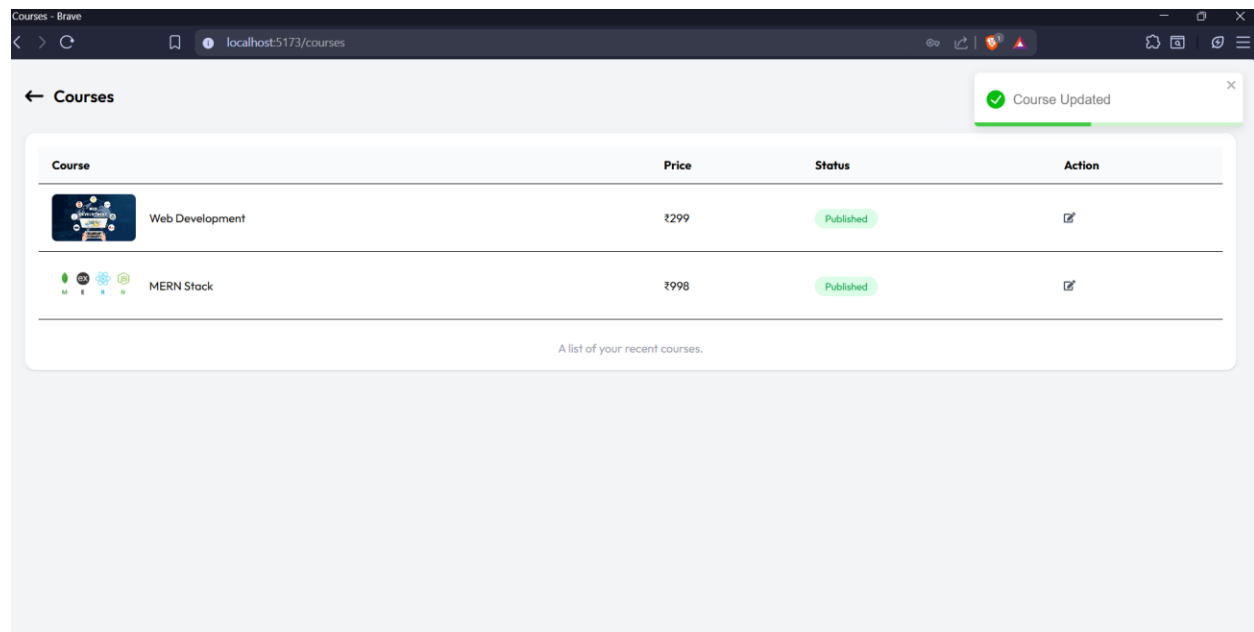












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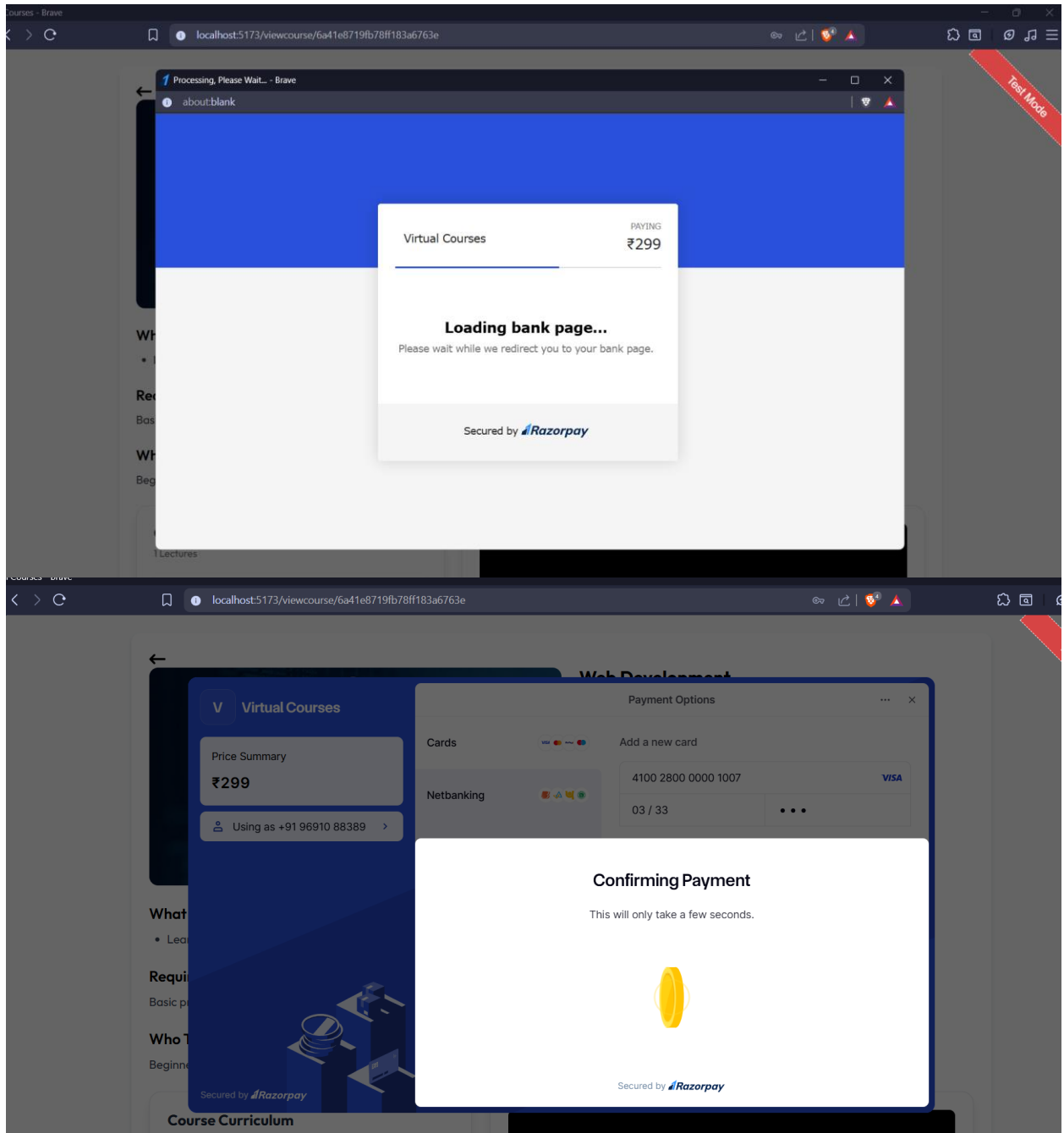
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